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Data Science

Blood Donation App

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Project Guide
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1. Introduction

- Problem Identified :
  - Difficulty in finding blood donors quickly during emergencies.
  - No centralized and secure system to search donors by blood group

- Solution Proposed :
  - Developed an Android –based Blood Donation App to connect donors and recipients easily.
  - Provides secure registration,blood group & notification.

2. Objectives

1. To enable blood search and matching functionality based on blood group to quickly connect donors with recipients.
2. To implement authentication and role management for donors, recipients, and admins to ensure secure access and data integrity.
3. To ensure data validation and privacy by applying proper input validation and secure data handling techniques.
4. To develop an admin module for managing donor records, blood requests, and user activities efficiently.

3. Scope

1. Can be applied in hospitals to quickly connect patients with available blood donors.
2. Can be useful during emergency situations to find donors instantly based on blood group.
3. Can be used by the general public to register as donors and accept blood request easily.
4. Can be helpful to improve fast and secure communication between donors and recipient.

4. Feature /Functionality

1. **Feature 1: User Authentication and Role Management**

   The system provides secure login and registration for different users such as donors, recipients, and admin, ensuring controlled access and proper role-based permissions.

2. **Feature 2: Blood Group-Based Matching**

   The system supports basic blood group matching by displaying donors with the same blood group, with scope for further enhancement.

4. Feature /Functionality

Feature 3: Blood Request Management System

Recipients can raise blood requests, and the system allows donors to view and respond to these requests, ensuring timely communication and support.

Feature 4: Admin Management System

An admin dashboard allows the administrator to manage donor records, blood requests, and user activities, ensuring proper monitoring and efficient system operation.

5. Outcome of Project

- User can register as a donor or recipient.
- User can log in securely using authentication.
- User can send blood requests during emergencies.
- Admin can manage users, donor records, and blood requests.

6. Technology Stack

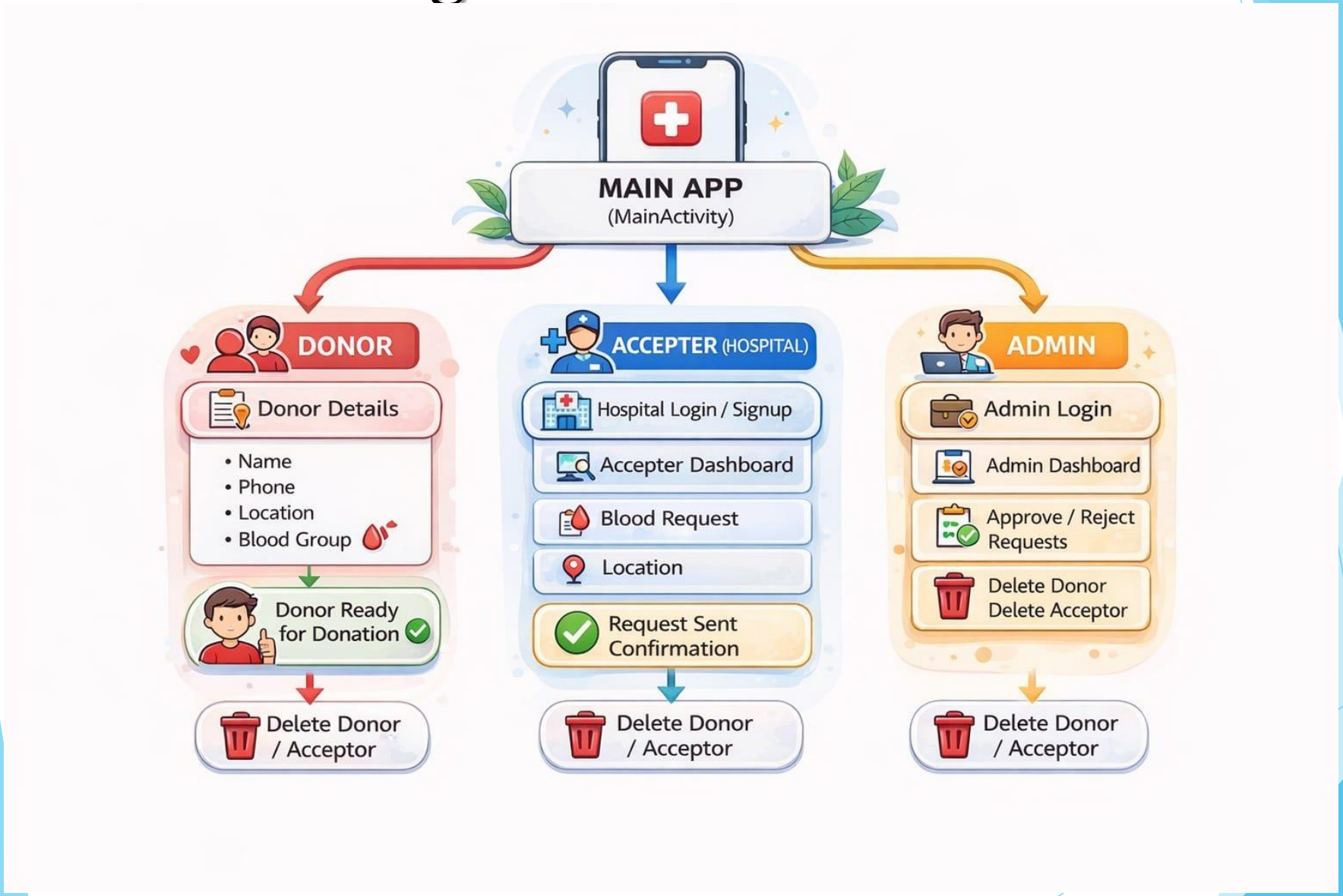
1. -Front End (GUI)

- The application will be developed using **Android Studio**.
- XML will be used to design the user interface pages like Login, Registration, and Request Blood.
- Java will be used for implementing the application logic and handling user interactions.

2.-Back End (Database)

- Firebase will be used as the backend database.
- Firebase real-time database will store user details such as name, blood group, location, and contact information.
- Firebase Authentication will manage secure login and user access control.

7. Block Diagram



Supporting SDG 3 through Blood Donation System

SDG 3 – Good Health and Well-being-Description:Ensure healthy lives and promote well-being for all at all ages.

Our Blood Donation App directly supports this goal because:

- It helps **connect donors and recipients quickly**
- It improves **access to blood during emergencies**
- It contributes to **saving lives and improving healthcare support**

Thank You...!!